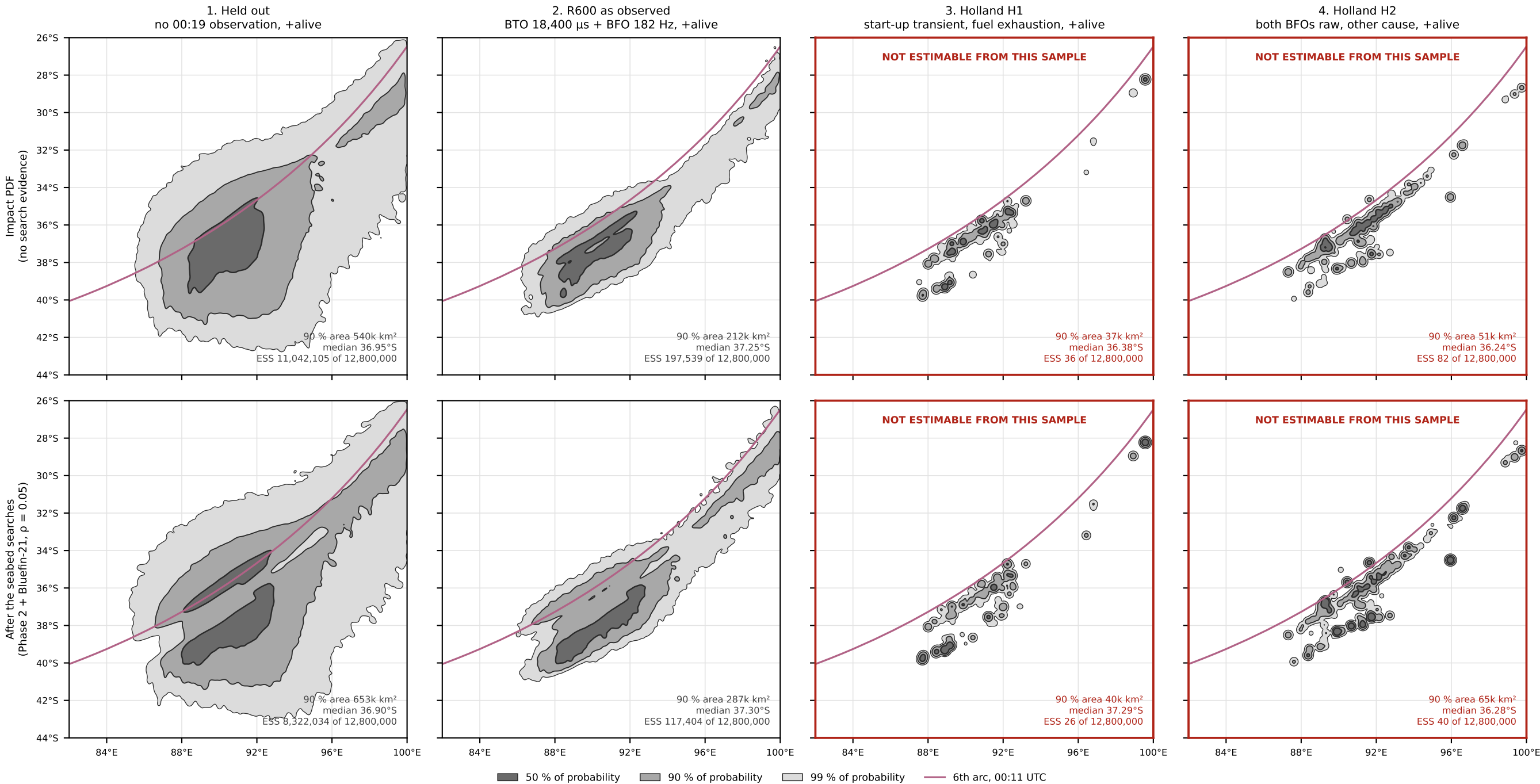




OBSERVATIONS engine/data/satcom-observations.csv, from the released unredacted SITA/Inmarsat logs. 00:19:29.416 UTC, R600 log-on request: BTO 18,400 μ s — the raw 23,000 μ s less the standard $-4,600$ μ s R600 log-on-channel correction — sd 63 μ s; BFO 182 Hz, tabulated sd 7 Hz. 00:19:37.443 UTC, R1200 log-on acknowledge: BFO -2 Hz, tabulated sd 7 Hz. Its BTO (raw 49,660 μ s, corrected by $4 \times 7,820$ μ s after Davey §5.2 to 18,380 μ s) is the anomalous value and is EXCLUDED from every panel. The constant BFO bias is Davey's 150 ± 25 Hz prior marginalised per particle, so the two 00:19 BFOs are scored JOINTLY and not as independent readings: $\ln L(\text{both}) - \ln L(\text{R600}) - \ln L(\text{R1200})$ is not a constant, ranging over $\approx 1,560$ nats across seed 1. The filter's effective BFO sd is 7.38 Hz (the tabulated 7 Hz variance plus a fixed 5.4 Hz²), recovered by exact reconstruction of the engine's log-likelihood ($R^2 = 1$).

PANELS, left to right. (1) No 00:19 observation at all — neither BFO and not the BTO arc; everything up to and including the 00:11 arc, with the fuel, dynamics and control model. (2) The 00:19:29 BTO and BFO at face value. (3) Holland's Hypothesis 1 (arXiv:1702.02432v3 §V, §VI): the SDU oscillator warming up after a power interruption, putting the acknowledge 17-130 Hz and the request a further 0-6 Hz above a steady oscillator — uniform, positive and shared between the bursts — paired with the fuel-exhaustion log-on cause, which also applies the §6 log-on lag density. (4) Holland's Hypothesis 2: some other log-on cause (software failure, loss of a critical SDU input, or attitude blocking the line of sight), so both BFOs at face value with no transient, and no lag density. Panel 2 $\rightarrow$ panel 4 is the increment from adding the 00:19:37 BFO under H2; panel 3 against panel 4 is Holland's two hypotheses. Holland himself used these bounds to bound the DESCENT RATE, not position; the ATSB took that result as the ± 25 NM corridor width and a descent kernel, never as a likelihood along the arc, so no panel here reproduces the ATSB's use.

LOWER ROW, the seabed-search evidence. ATSB Phase 2 union 120,486.5 km² (deep-tow side-scan, GO Phoenix and Dong Hai Jiu SAS, AUV side-scan) plus Bluefin-21/Artemis 771.4 km²; coverage rasterised at 0.01°; detection probability $q = 0.945$ Phase 2 and 0.900 Bluefin-21, conditional on a detectable target; undetectable fraction $p = 0.05$; point target — the size response $g(W)$ is not yet implemented; shared miss dependence where campaigns overlap. Ocean Infinity 2018 and 2025-26 are NOT included. PRIOR run eof-289-full: 289.7° initial track at 18:01:49 UTC (Davey Fig. 4.2), 4 seeds $\times 3.2 \times 10^6$ impacts. Bands are 50/90/99 % highest-posterior-density regions on a 0.02° grid smoothed at 0.1° (6 NM); areas on the authalic sphere. **+alive**: end of flight's declared existence constraint, PROVISIONAL-OVERNIGHT, that the aircraft was airborne at 00:19:37.443 so that the log-on acknowledge could be sent at all — a datum separate from that burst's BTO and BFO values. It binds only on the held-out arm, which otherwise puts 10.2 % of its weight before that burst.

CONVERGENCE Each panel is an importance-weighted reading of the SAME 12.8×10^6 impacts, which were not drawn with the 00:19 bursts in hand, so a sharp 00:19 likelihood collapses the weights. ESS is the Kish effective sample size of those weights. A panel below 1,000 effective impacts is labelled NOT ESTIMABLE: its bands are the few surviving particles, not a posterior, and its area, median and evidence are reported as unconverged, not as results. Fixing it needs a proposal that already carries the 00:19 data, which is end of flight's to build, not a longer run of this one.